Resume Analyser

Resume Analysis and Job Compatibility Prediction Using Machine Learning

Subject: Machine Learning

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Problem statement

- **Too many resumes.** A single opening can attract hundreds of applications, and screening them by hand is slow and inconsistent.
- **Keyword filters are brittle.** A filter looking for "budgeting" misses a candidate who wrote "financial planning".
- **Candidates get no feedback.** A rejection does not say which required skills or experience were missing.
- **Scores are hard to trust.** Many screening tools return a number with no explanation of how it was reached.

Proposed solution

Resume Analyser reads a resume and a job description using NLP and machine learning, and returns:

- **Role prediction:** which of 24 job categories the resume fits best, with a confidence.
- **Compatibility score:** a calibrated probability that the resume fits the job.
- **Candidate ranking:** an ordered shortlist when several resumes are uploaded for one job.
- **Explanation:** what raised or lowered the score, and which required skills are missing.

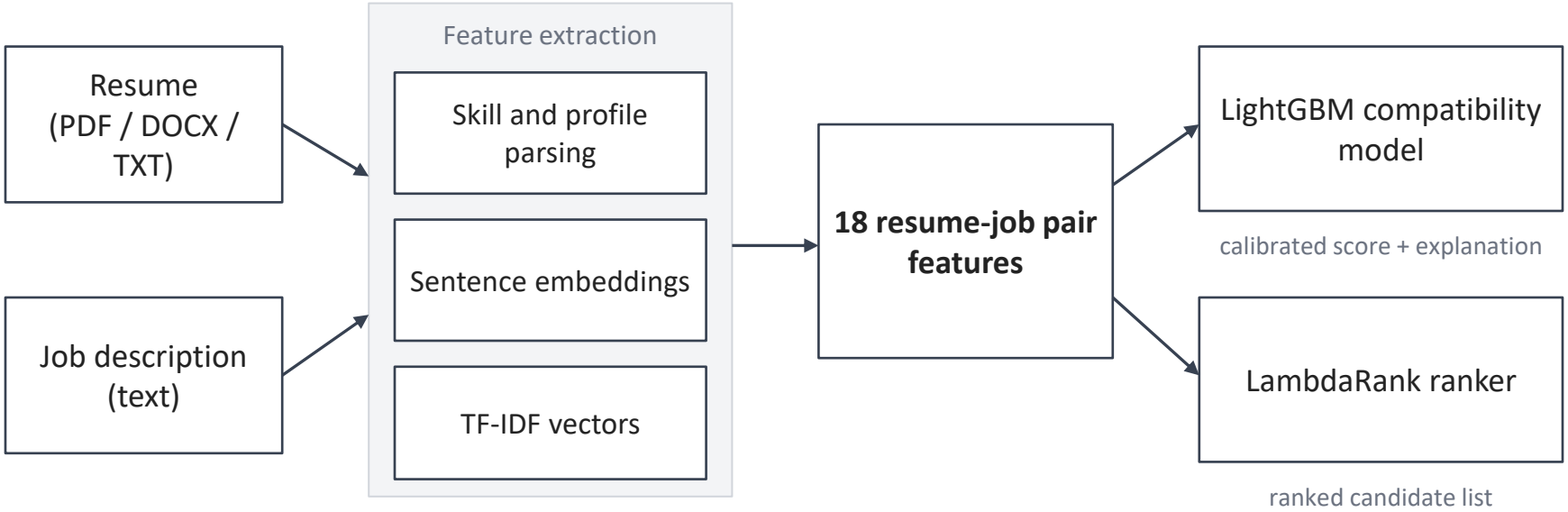
Research objectives covered

1. **Resume parsing:** extract skills, education, experience and certifications.
2. **Semantic matching:** compare resume and job meaning with sentence embeddings, not only keywords.
3. **Compatibility prediction:** train a supervised model on resume-job pairs.
4. **Candidate ranking:** learn to order candidates for a job (learning to rank).
5. **Explainability:** show which features drove each score.
6. **Evaluation:** compare against baselines; run ablation, calibration and a bias check.

Methodology

1. **Collect data:** Kaggle resume dataset, cleaned and de-duplicated.
2. **Parse resumes:** mine a skill list from the resumes' own Skills sections; extract education and years of experience.
3. **Create training pairs:** generate job descriptions and graded relevance labels (0 to 3).
4. **Engineer features:** 18 numeric features for every resume-job pair.
5. **Train and compare:** baselines, logistic regression, LightGBM classifier and ranker; calibrate probabilities.
6. **Explain and deploy:** TreeSHAP contributions and a Flask web app with an HTML/CSS/JavaScript frontend.

Pipeline



Separately, a TF-IDF + Linear SVM classifier predicts the resume's role (24 categories).

Tools and technologies

Area	Tools
Language	Python
Machine learning	scikit-learn, LightGBM
NLP	TF-IDF, Sentence-Transformers (all-MiniLM-L6-v2)
Data handling	Pandas, NumPy, SciPy, joblib
Resume parsing	pypdf, python-docx, regular expressions
Web app	Flask (REST API), HTML, CSS, JavaScript
Dataset access	KaggleHub

About the dataset

- **Source:** Kaggle Resume Dataset (snehaanbhawal).
- **Size:** 2,481 unique resumes after removing duplicates.
- **Labels:** 24 job categories, e.g. Information Technology, HR, Finance, Engineering, Healthcare, Designer, Sales, Aviation.
- **Content:** plain resume text with sections such as Summary, Skills, Experience and Education.
- **Missing pieces:** the dataset has no job descriptions and no resume-job match labels, so we created them (next slide).

Creating training pairs (weak supervision)

Example job description (same format as the generated ones)

Finance Analyst
Requirements: 3+ years of relevant experience, a bachelor's degree. Strong skills in accounting, budgeting, financial reporting and auditing. Familiarity with accounts payable is a plus.

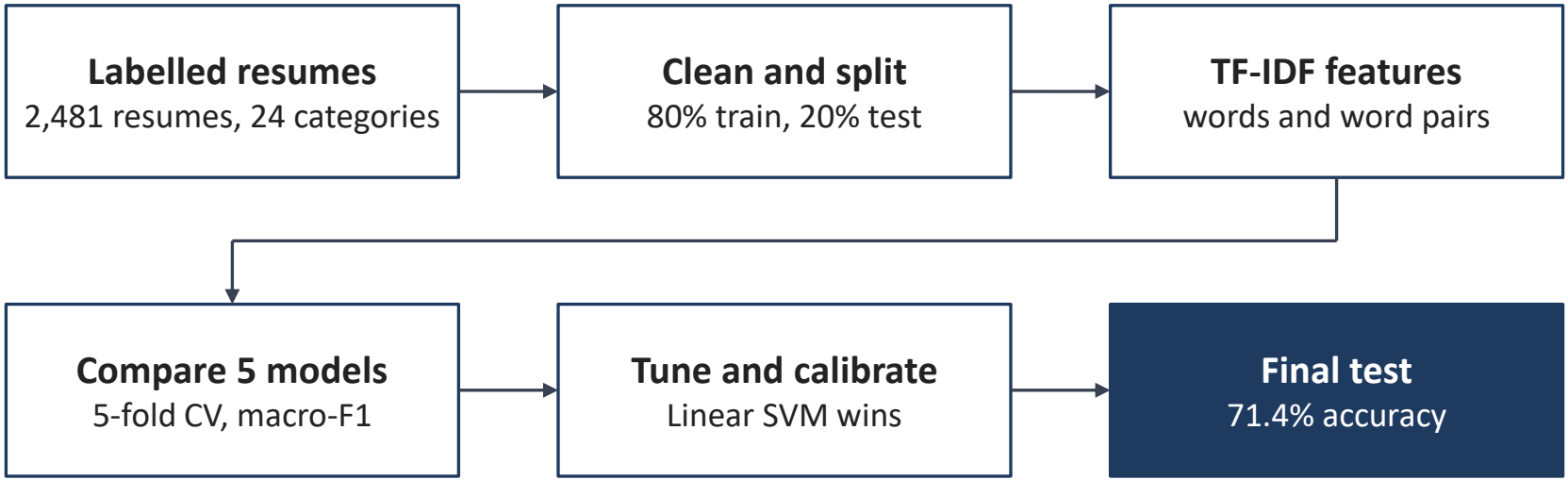
- Skills in each job description are mined from training resumes of one category
- Train and test jobs use different skill pools
- Resumes are split by candidate (80 / 20)

Relevance label of a resume-job pair

Label	Meaning
3	Same category
2	Closely related (e.g. Accountant and Finance)
1	Same broad family
0	Unrelated

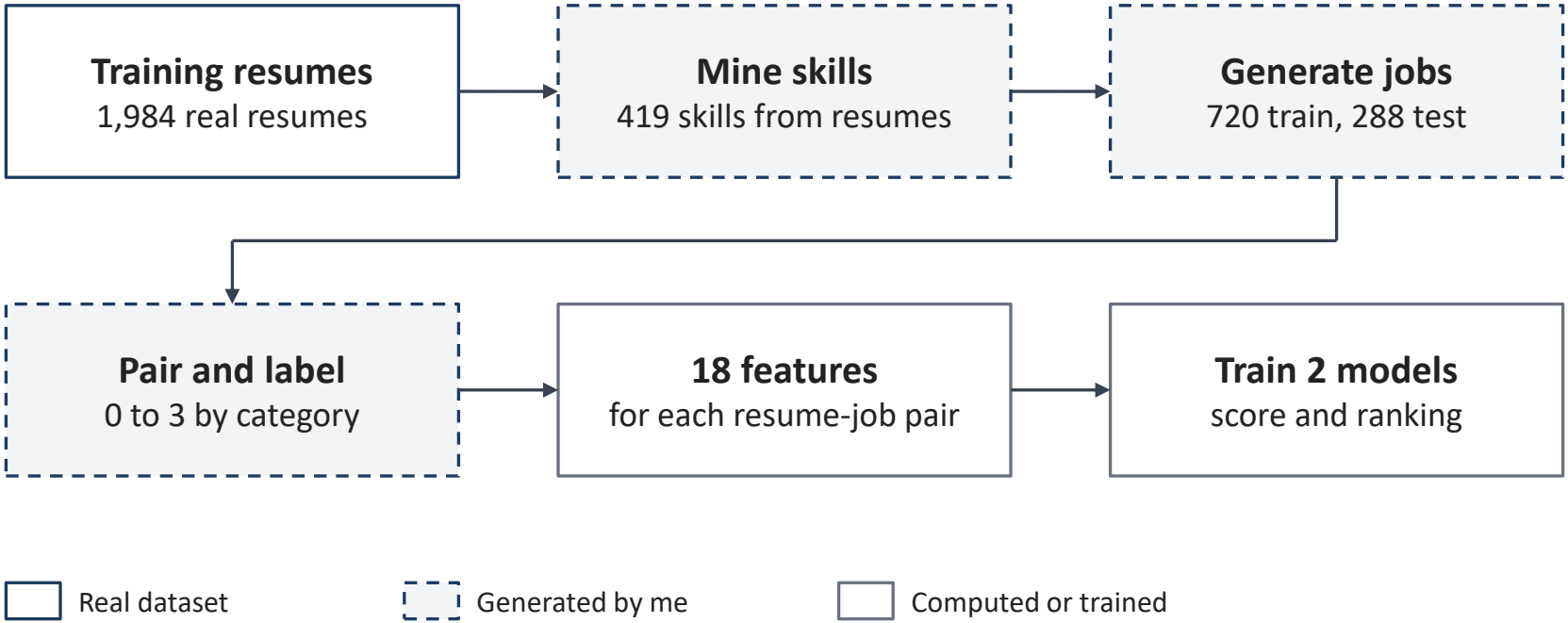
"Compatible" means label 2 or 3.
28,800 training pairs, 11,520 test pairs.

How the role classifier is trained



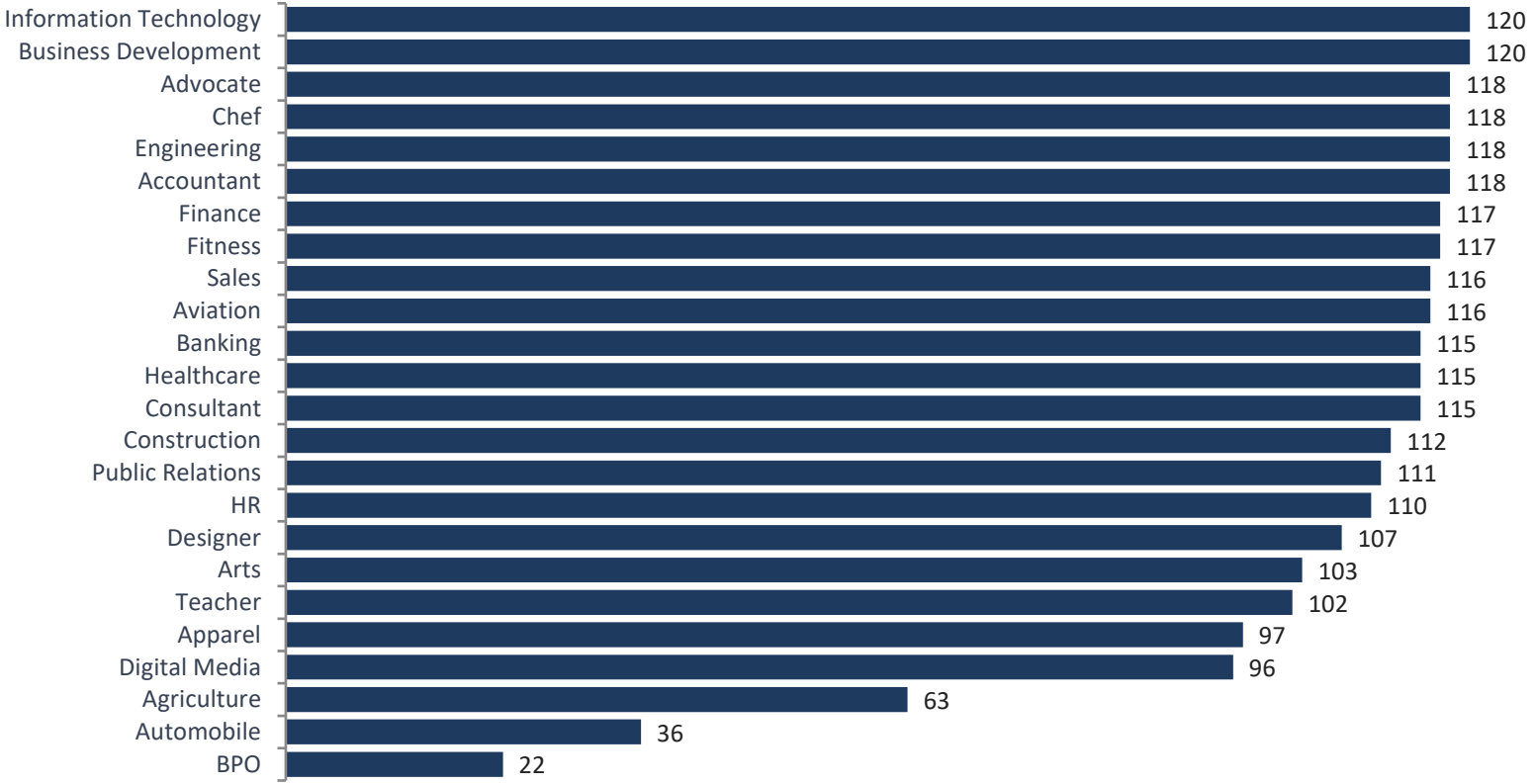
Every step here uses the dataset's real labels. The test set is touched only once, at the end.

How the compatibility model is trained

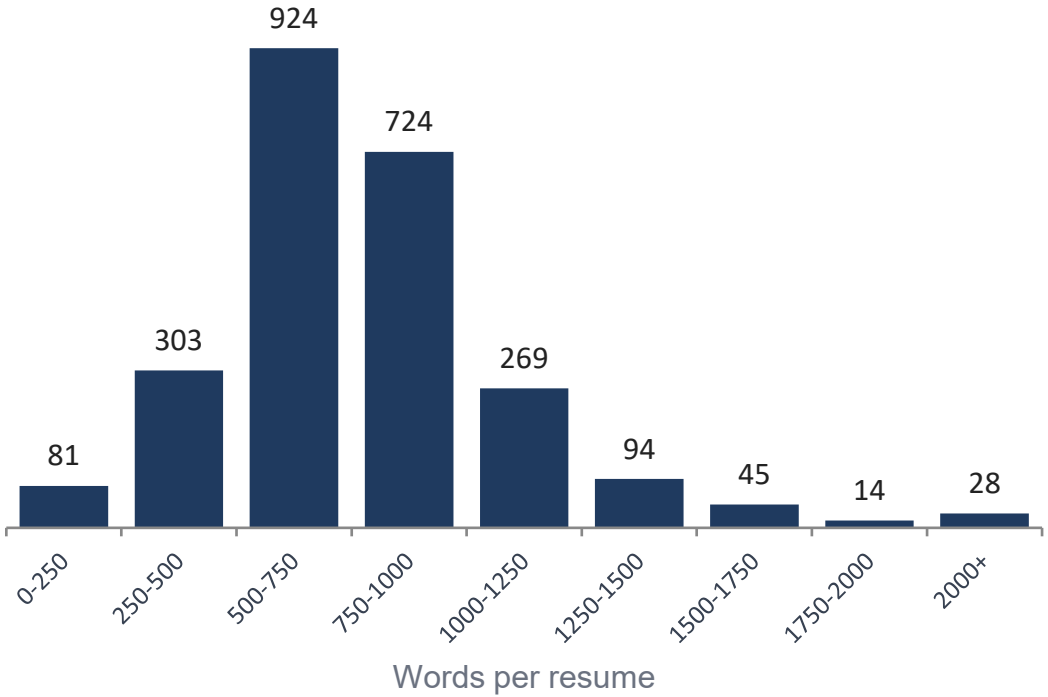


Then LightGBM is calibrated (isotonic) on separate jobs, LambdaRank orders candidates per job, and TreeSHAP explains each score.

EDA: resumes per category



EDA: resume length



- Average length is about 790 words; most resumes have 250 to 1,000 words
- Some run past 2,000 words, more than a sentence model reads at once
- Resumes have a clear structure: Summary, Skills, Experience, Education
- The Skills sections became our skill list

Feature engineering: 18 features

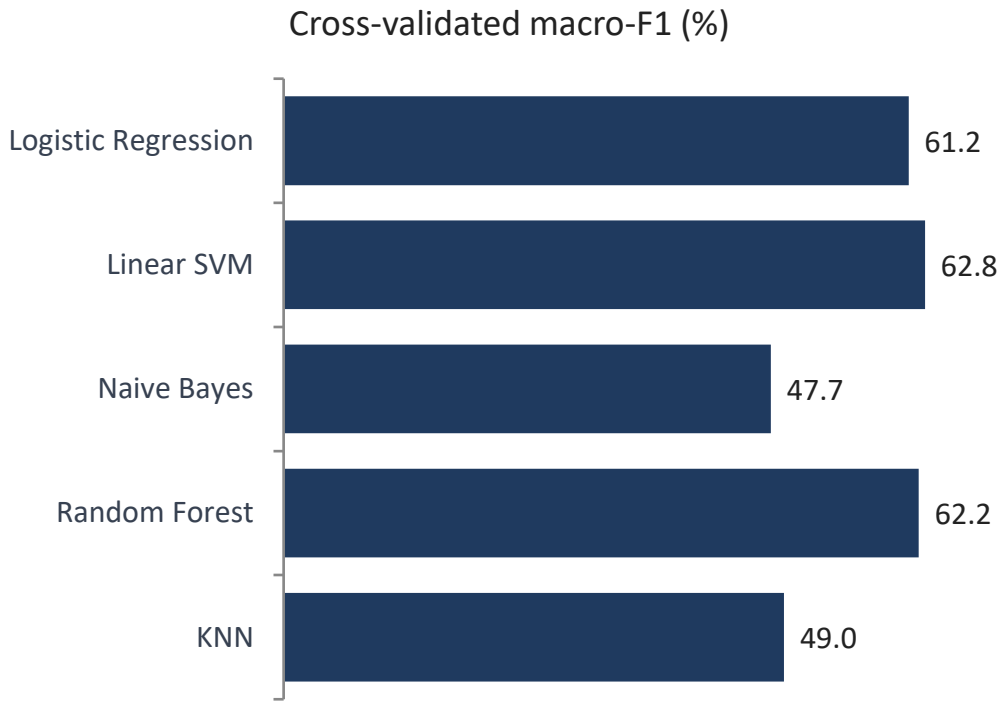
Group	Features
Semantic	Embedding cosine similarity of resume and job
Keyword	TF-IDF cosine similarity
Skills	Required-skill coverage, matched and missing counts, Jaccard overlap, number of skills on each side
Experience	Candidate years, required years, ratio, gap, meets requirement (yes/no)
Education	Candidate level, required level, gap, meets requirement (yes/no)

Skills come from a 419-skill list mined from the resumes' own Skills sections. Experience is read from stated years and dated job ranges; education level from degree keywords.

NLP techniques used

- **Text cleaning:** lowercasing, tokenization, stop-word removal.
- **TF-IDF:** unigrams and bigrams with sublinear term frequency; used for the role classifier and as a similarity feature.
- **Sentence embeddings:** all-MiniLM-L6-v2; each resume is split into chunks and the chunk vectors are averaged.
- **Skill extraction:** longest-match lookup against the mined skill list.
- **Profile parsing:** education level, years of experience and certifications with rules and regular expressions.
- **Keyword gaps:** important job terms (by TF-IDF weight) that never appear in the resume.

Results: role classifier



- **Best model:** Linear SVM on TF-IDF (tuned with GridSearchCV).
- **Test accuracy:** 71.4%, macro-F1 0.67 on held-out resumes.
- **Strongest classes:** Designer, Information Technology and HR (F1 about 0.9).

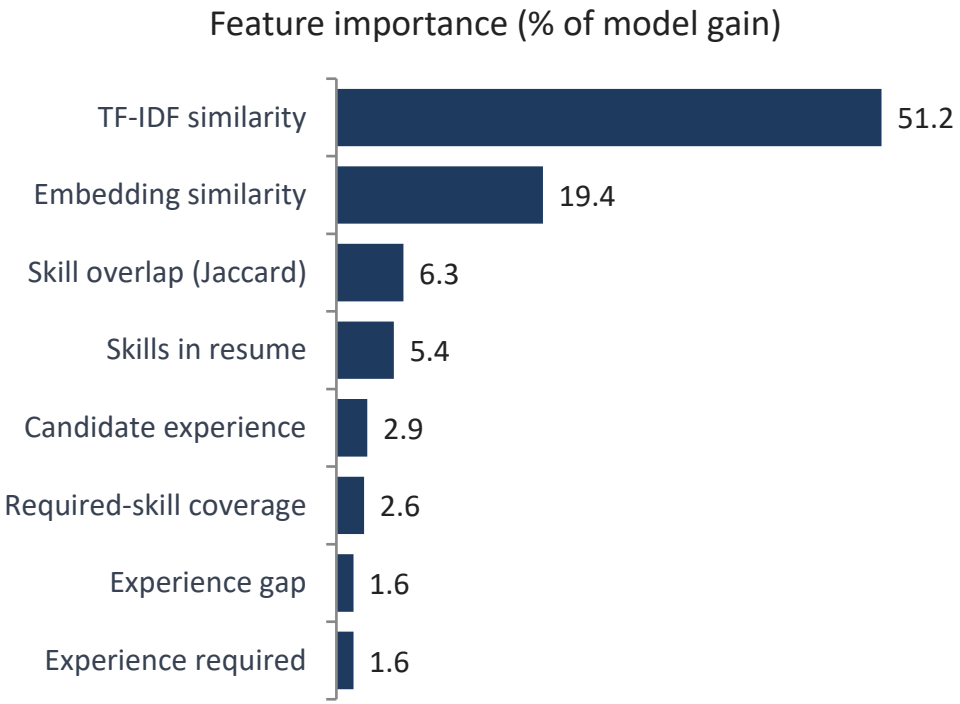
Results: compatibility and ranking

Model	ROC-AUC	NDCG@10
Keyword overlap (baseline)	0.694	0.645
TF-IDF cosine (baseline)	0.769	0.643
Embedding cosine (baseline)	0.759	0.626
Logistic Regression	0.807	0.670
LightGBM classifier (calibrated)	**0.804**	**0.699**
LambdaRank ranker	0.802	0.677

- ROC-AUC 0.80 vs 0.77 for the best baseline
- NDCG@10 of 0.70: relevant candidates rank near the top
- Calibrated probabilities (expected calibration error 0.018)
- Test candidates and test jobs were never seen in training

Evaluated on synthetic job descriptions with category-derived relevance labels; test candidates and jobs were not seen in training.

Importance of each feature



Ablation study

Group removed	ROC-AUC
None (full model)	**0.806**
Embeddings	0.779
TF-IDF	0.789
Skills	0.795
Experience	0.804
Education	0.803

Evaluated on synthetic job descriptions with category-derived relevance labels; test candidates and jobs were not seen in training.

Explainability in action

Finance resume vs Finance Analyst job: compatibility 88%

Helped the score	Impact
Keyword similarity (TF-IDF), 0.056	+1.77
Semantic similarity (embeddings), 0.50	+0.63
Only 3 required skills missing	+0.29
Required-skill coverage, 57%	+0.23

Hurt the score	Impact
Job requires 3 years of experience	-0.32
Candidate's 20 years of experience	-0.22
49 skills found in the resume	-0.17
Experience surplus of 17 years	-0.01

- Positive impact pushes the model toward "compatible"; negative pushes away
- The dominant signal is text similarity; experience features make small adjustments
- Candidates and recruiters can see exactly why a score was given

Error analysis and fairness check

Classifier: weakest and strongest classes

Class	Resumes	F1
BPO	4	0.00
Apparel	19	0.33
Automobile	7	0.44
Designer	21	0.91
Information Technology	24	0.89

Bias check: change in compatibility score

Identity text added	Avg change	Flips
female name	1.0%	2 / 120
male name	1.0%	1 / 120
south-asian name	0.8%	2 / 120
east-asian name	0.5%	1 / 120
birth year 1962	0.6%	2 / 120

- Low F1 goes with small classes (BPO has only 22 resumes in total)
- Compatibility mistakes mostly involve neighbouring fields, e.g. Public Relations vs Arts
- Names, pronouns and birth year shift scores by about 1% on average; protected attributes are never inputs

Input and output

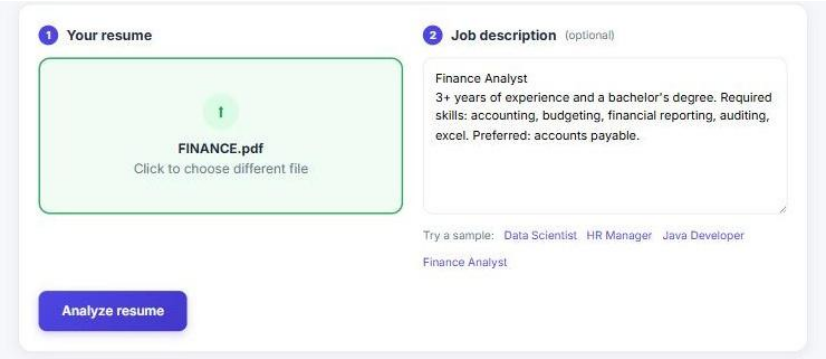
Input

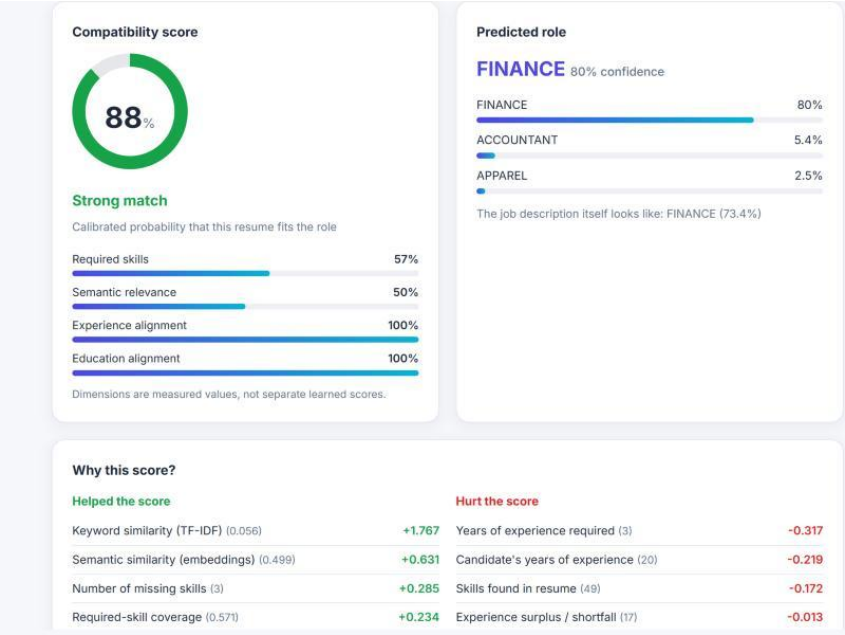
- Resume file (PDF, DOCX or TXT)
- Job description text
- Or 2 to 30 resumes to rank for one job

Output

- Predicted role with confidence
- Compatibility score (%) with four dimensions: required skills, semantic relevance, experience, education
- Matched and missing skills and keywords
- "Why this score?": features that helped and hurt
- Ranked list of candidates

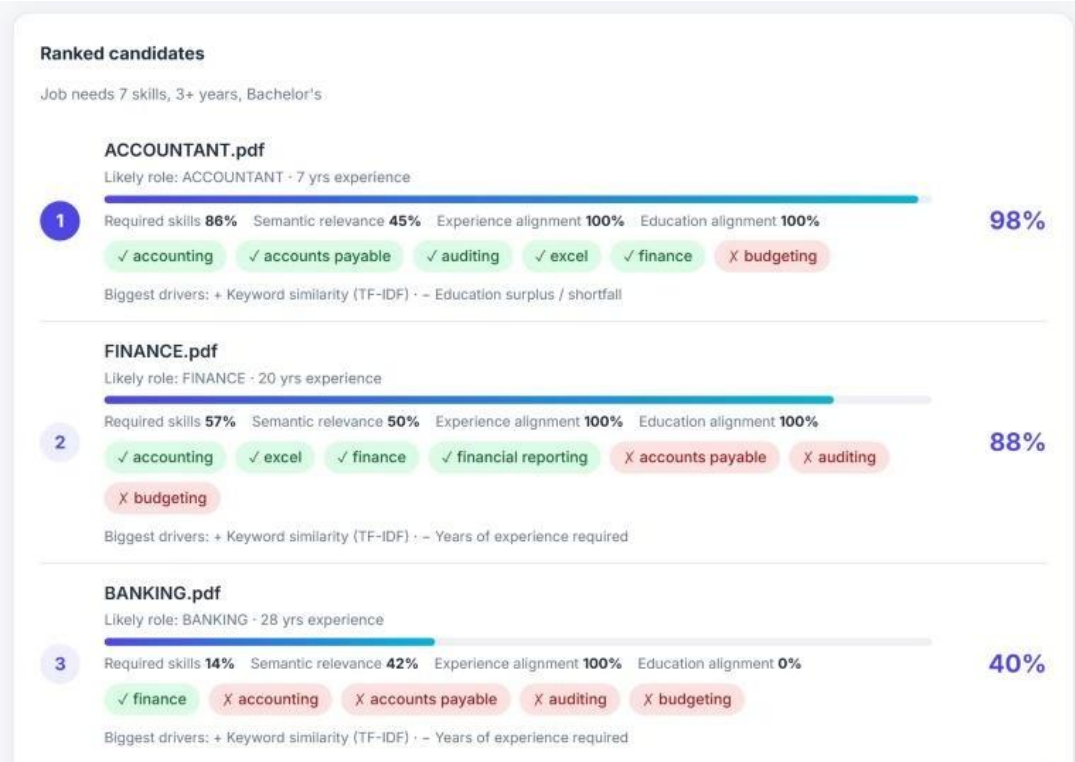
Demo: analyze a resume





- Upload a PDF, DOCX or TXT resume
- Paste a job description or pick a sample
- The score, predicted role and explanation appear in a few seconds

Demo: rank candidates



- Five resumes ranked against one finance job
- Accountant 98%, Finance 88%, Banking 40%
- IT (7%) and Chef (4%) are ranked last
- Each candidate shows matched and missing skills

Unique selling points

- **Explainable:** every score comes with the features that raised or lowered it (TreeSHAP).
- **Semantic:** embeddings capture meaning, so different wording of the same skill still matches.
- **Calibrated:** scores are probabilities, checked against observed outcomes.
- **Ranking built in:** a LambdaRank model orders candidates for a job.
- **Self-built skill list:** 419 skills mined from real resumes, not typed by hand.
- **Properly evaluated:** baselines, ablation, error analysis and a bias check.
- **Fairness-aware:** no protected attributes are used; adding names or pronouns changed scores by about 1% on average.

Future scope

- **Real hiring data:** train on recruiter-labelled resume-job pairs and hiring outcomes.
- **Real job postings:** replace generated descriptions with large public job-posting datasets.
- **Fine-tuned models:** cross-encoders or resume-specific transformers.
- **Richer parsing:** named-entity recognition for titles and projects; OCR for scanned PDFs.
- **Skill ontology:** link to ESCO or O*NET to cover niche and emerging skills.
- **Fairness tooling:** name masking, formal bias audits and human review.
- **Smarter thresholds:** tune the match cutoff per role to catch more suitable candidates.
- **Deployment:** cloud hosting, user accounts, feedback loop and periodic retraining.

Conclusion

- **End to end:** a working system that parses resumes, matches them semantically, predicts compatibility, ranks candidates and explains each score.
- **Role classifier:** 71.4% accuracy across 24 categories, about 17 times better than random guessing.
- **Compatibility model:** beats keyword, TF-IDF and embedding baselines (ROC-AUC 0.80 vs 0.77, NDCG@10 0.70).
- **Trustworthy outputs:** calibrated probabilities, TreeSHAP explanations, ablation, error analysis and a bias check.
- **Next step:** validate and retrain on recruiter-labelled resume-job pairs.

References

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Thank you

https://github.com/Kiwi-Supreme/Resume-Analyser